Calculus and Linear Algebra

The Chain Rule

Composite functions and nested derivatives

Lecture 18

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Today's plan: intentionally light

First part of the class

Complete Lecture 17: linearity, power rule, product rule, quotient rule, tangent lines and exercises.

Then we add only one major new idea:

How do we differentiate a function inside another function?

- Recognise composite functions.
- State and use the chain rule.
- Combine the chain rule with rules already known.

Recall: composition of functions

If

$$u = g(x), \qquad y = f(u),$$

then

$$y = f(g(x)) = (f \circ g)(x).$$

Example:

$$y = (3x - 1)^5.$$

Think of it as

$$\boxed{x \xrightarrow{g} u = 3x - 1 \xrightarrow{f} y = u^5.}$$

The derivative must account for both stages.

What is the outer function? What is the inner function?

For each expression, identify the two layers.

Expression	Inner $g(x)$	Outer $f(u)$
$(x^2+1)^4$	x^2+1	u^4
$\sqrt{3x-2}$	$3x-2$	$\sqrt{u}$
$\dfrac{1}{(2x+5)^3}$	$2x+5$	u^{-3}
$\left(1+(x-1)^2\right)^5$	$1+(x-1)^2$	u^5

Habit to build

Before differentiating, first identify the layers.

The chain rule

Theorem If g is differentiable at x and f is differentiable at $g(x)$, then

$$\boxed{\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x).}$$

Using $u = g(x)$:

$$\boxed{\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.}$$

In words

Differentiate the **outer** function, keep the inner expression unchanged, then multiply by the derivative of the **inner** function.

Why the extra factor matters

Compare

$$y = x^5 \qquad \Longrightarrow \qquad y' = 5x^4$$

with

$$y = (3x - 1)^5.$$

Differentiating only the outer power would give an incomplete answer:

$$5(3x - 1)^4.$$

The inner function has derivative 3, so

$$\boxed{y' = 5(3x - 1)^4 \cdot 3 = 15(3x - 1)^4.}$$

Power of a function

A particularly useful form of the chain rule is

$$\boxed{\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x).}$$

Example:

$$y = (x^2 + 4)^3.$$

Then

$$y' = 3(x^2 + 4)^2 \cdot 2x,$$

so

$$\boxed{y' = 6x(x^2 + 4)^2.}$$

Example: a square root

Differentiate

$$y = \sqrt{x^2 + 4} = (x^2 + 4)^{1/2}.$$

Outer derivative:

$$\frac{1}{2}(x^2 + 4)^{-1/2}.$$

Inner derivative:

$$2x.$$

Therefore

$$\boxed{y' = \frac{x}{\sqrt{x^2 + 4}}.}$$

Example: a reciprocal power

Differentiate

$$y = \frac{1}{(x+1)^2} = (x+1)^{-2}.$$

By the chain rule,

$$y' = -2(x+1)^{-3} \cdot 1.$$

Hence

$$\boxed{y' = -\frac{2}{(x+1)^3}}, \qquad x \neq -1.$$

Notice that differentiation does not change the original domain restriction.

Nested functions: more than two layers

Consider

$$y = (1 + (2x - 3)^2)^4.$$

There are three layers:

$$x \longmapsto 2x - 3 \longmapsto 1 + (2x - 3)^2 \longmapsto (1 + (2x - 3)^2)^4.$$

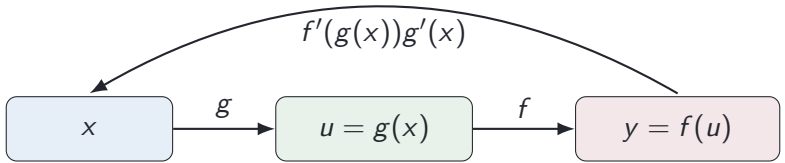
Differentiate from the outside inward:

$$y' = 4(1 + (2x - 3)^2)^3 \cdot 2(2x - 3) \cdot 2.$$

Thus

$$\boxed{y' = 16(2x - 3)(1 + (2x - 3)^2)^3.}$$

A useful visual workflow



1. Identify the outer function.
2. Differentiate it, leaving the inner expression untouched.
3. Multiply by the derivative of the inner expression.
4. Repeat if another layer remains.

Chain rule combined with the product rule

Differentiate

$$f(x) = x^2(x+1)^3.$$

This is a product, and one factor is composite:

$$f'(x) = 2x(x+1)^3 + x^2 \cdot 3(x+1)^2.$$

Factor if useful:

$$\boxed{f'(x) = x(x+1)^2(5x+2).}$$

Strategy

First identify the **outermost algebraic operation**: here it is a product. Then differentiate each component correctly.

Chain rule combined with the quotient rule

Differentiate

$$f(x) = \frac{(x^2+1)^2}{x-1}, \qquad x \neq 1.$$

Let

$$N(x) = (x^2+1)^2, \qquad D(x) = x-1.$$

Then

$$N'(x) = 2(x^2+1)(2x) = 4x(x^2+1), \qquad D'(x) = 1.$$

Therefore

$$\boxed{f'(x) = \frac{4x(x^2+1)(x-1) - (x^2+1)^2}{(x-1)^2}.}$$

Common mistakes

- Forgetting the derivative of the inner function:

$$((2x+1)^4)' \neq 4(2x+1)^3.$$

- Differentiating the inner function too early and losing the outer structure.
- Confusing product rule and chain rule:

$$(x^2+1)^3 \text{ is a composition, not a product.}$$

- Expanding a complicated power unnecessarily before differentiating.

Good practice

Write one line for the structure before doing algebra.

Exercise set

Exercise Differentiate.

1. $f(x) = (4x - 3)^6$.
2. $g(x) = \sqrt{x^2 + 1}$.
3. $h(x) = \dfrac{1}{(3x + 2)^4}$.
4. $p(x) = \left(1 + (x - 2)^2\right)^3$.
5. $q(x) = x(x^2 + 1)^4$.

For each function, identify the outer and inner functions before computing the derivative.

Exercise set: solutions I

Solution

$$\boxed{f'(x) = 24(4x-3)^5.}$$

$$\boxed{g'(x) = \frac{x}{\sqrt{x^2+1}}.}$$

$$\boxed{h'(x) = -\frac{12}{(3x+2)^5}.}$$

Each answer contains the derivative of the inner function as a factor.

Exercise set: solutions II

Solution

For

$$p(x) = (1 + (x-2)^2)^3,$$

$$\boxed{p'(x) = 6(x-2)(1 + (x-2)^2)^2.}$$

For

$$q(x) = x(x^2+1)^4,$$

product rule plus chain rule gives

$$\boxed{q'(x) = (x^2+1)^4 + 8x^2(x^2+1)^3.}$$

Tangent line to a composite function

Let

$$f(x) = (x^2 + 1)^2.$$

Find the tangent line at $x_0 = 1$.

$$f(1) = 4.$$

By the chain rule,

$$f'(x) = 2(x^2 + 1)(2x) = 4x(x^2 + 1),$$

so

$$f'(1) = 8.$$

Hence

$$\boxed{y - 4 = 8(x - 1)}, \qquad \boxed{y = 8x - 4.}$$

Concept check

Exercise Which derivative is correct for

$$f(x) = (x^2 - 3)^5?$$

1. $5(x^2 - 3)^4$.
2. $10x(x^2 - 3)^4$.
3. $5x(x^2 - 3)^4$.
4. $10x(x^2 - 3)^5$.

Solution The correct answer is

$$\boxed{10x(x^2 - 3)^4.}$$

What we add today

After completing Lecture 17, the new rule is

$$\boxed{(f \circ g)'(x) = f'(g(x))g'(x).}$$

It allows us to differentiate nested expressions without expanding them. We can now combine:

linearity, power, product, quotient, chain rule.

Looking ahead

Next lecture we complete the derivative toolbox with

$$\boxed{e^x,\ \ln x,\ \sin x,\ \cos x,\ \tan x.}$$

The chain rule will immediately give derivatives such as

$$e^{x^2}, \qquad \ln(1+x^2), \qquad \sin(3x-1).$$

We will also introduce higher-order derivatives and logarithmic differentiation.